PC02
2202436H

Deep Learning Car Damage Verification



Presented by Yong Man Ting

VEHICLE DAMAGE

Our goal is to develop a robust and high performance model for classifying an image of a car into different types of damages automatically with the help of computer vision techniques.

By accurately identifying the damages, the insurance company can assess the legitimacy of the claim and make informed decisions regarding payouts.



CHALLENGE

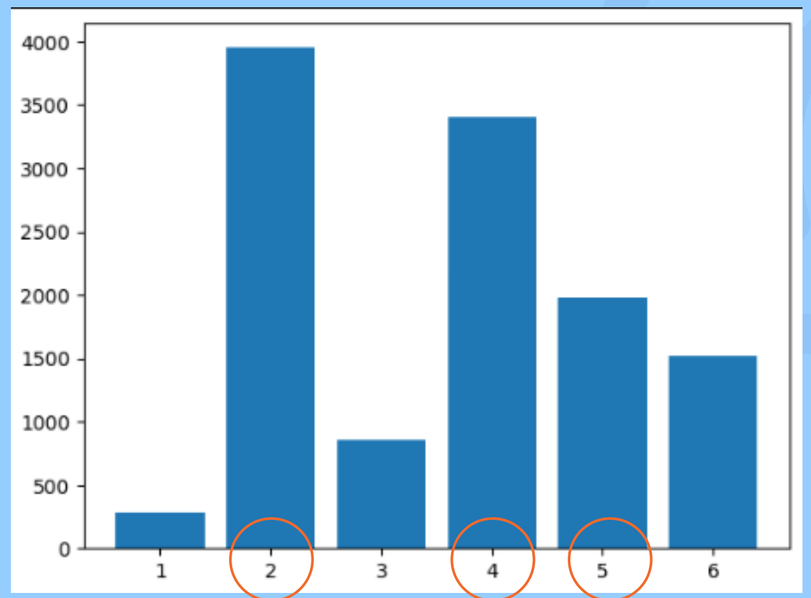
In the insurance industry, processing claims for vehicle damage is a common and most essential task. With the advancement in AI and Computer Vision, the users can settle the claims online instantly by uploading the images of the damaged car with the insurance company.

Now, insurance companies face the constant challenge of identifying fraudulent claims. It's a common practice for the users to submit the fraudulent images as a part of the claim settlement process. This brings out the threat/challenge to the insurance companies to identify the fraudulent claims which leads to significant financial losses.

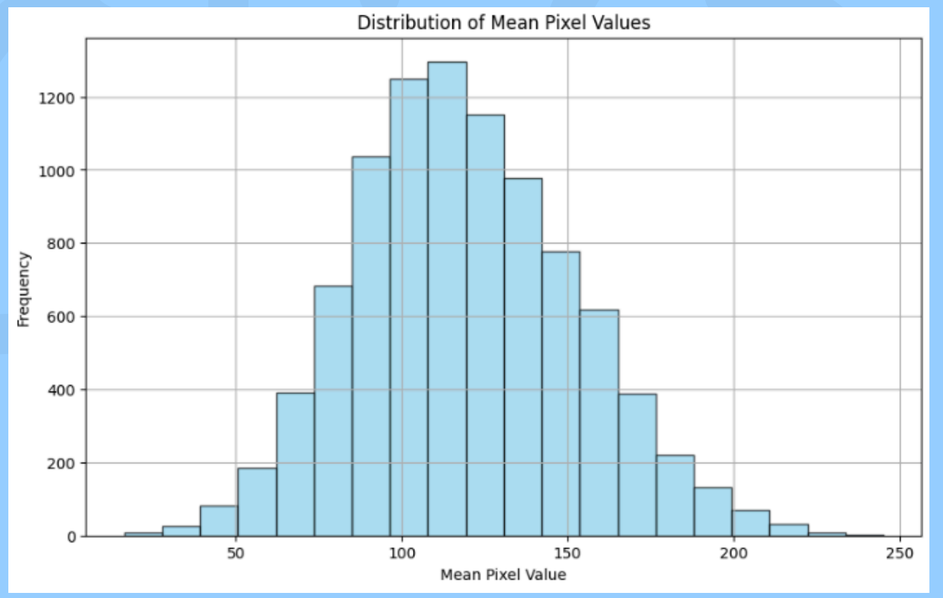
Fraudulent claims often involve exaggerating the extent of damage or submitting false claims altogether. In this problem, we will focus on the first type of problem i.e. exaggerating the extent of damage. To mitigate these losses and maintain the integrity of their operations, insurance companies must develop effective methods to flag out these claims most accurately and efficiently.

DATASET

This dataset is found from Kaggle.
There were 6 classes initially, however, due to the imbalance distribution, I decided to only pick the top 3 classes for this model.



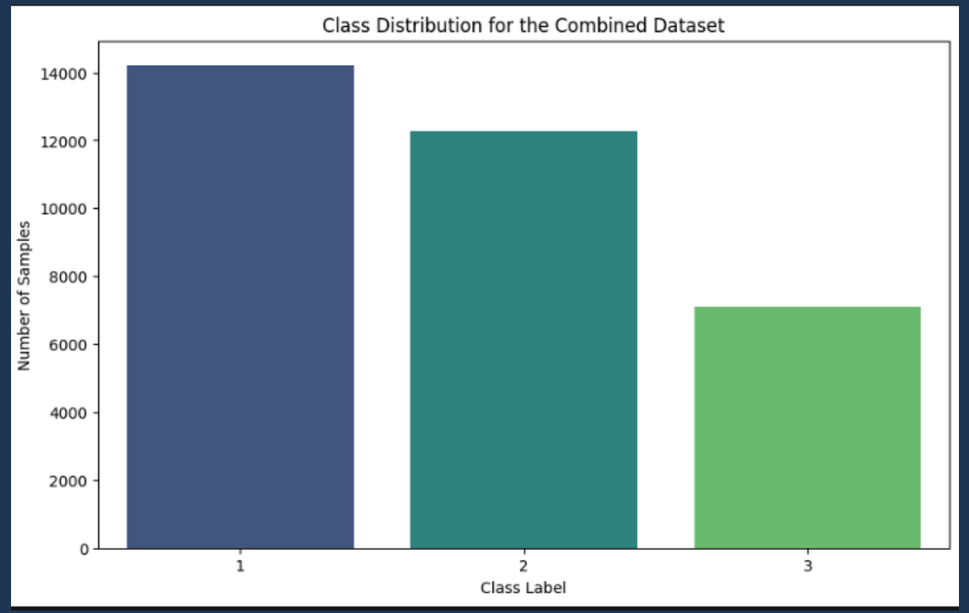
Scratch Dent Glass Shatter

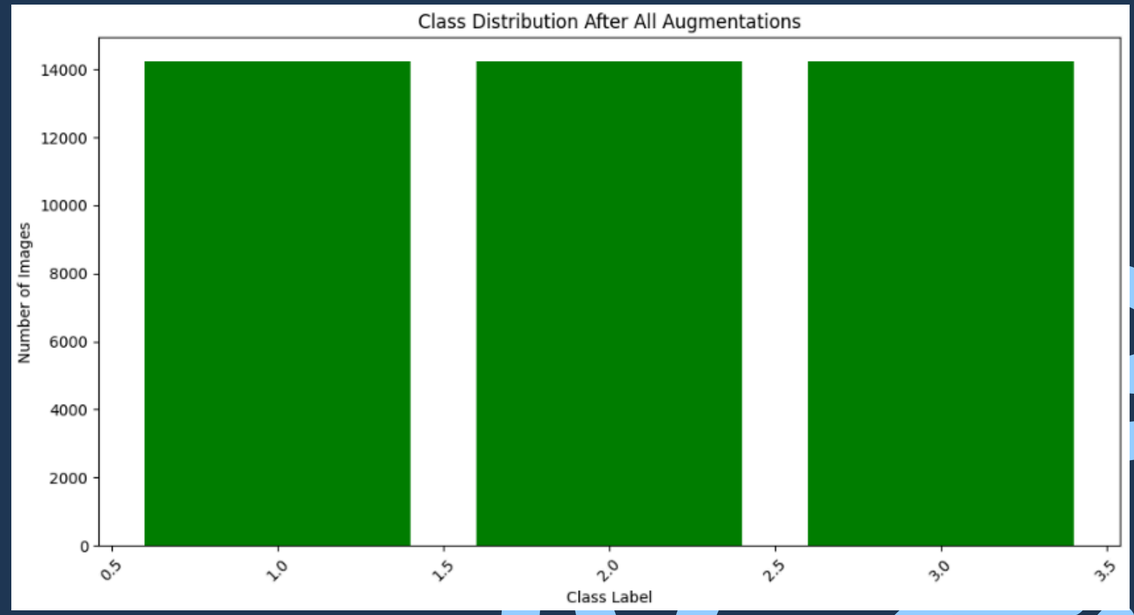



```
The size of 9782.jpg is: 811 x 608 pixels
The size of 2508.jpg is: 1376 x 1032 pixels
The size of 1734.jpg is: 833 x 624 pixels
The size of 11650.jpg is: 1230 x 922 pixels
The size of 6375.jpg is: 1413 x 1059 pixels
The size of 5786.jpg is: 847 x 635 pixels
The size of 10099.jpg is: 801 x 600 pixels
The size of 6992.jpg is: 1304 x 978 pixels
The size of 4548.jpg is: 1146 x 859 pixels
The size of 2021.jpg is: 818 x 613 pixels
The size of 488.jpg is: 900 x 675 pixels
The size of 7226.jpg is: 838 x 628 pixels
The size of 8973.jpg is: 1487 x 1115 pixels
The size of 9100.jpg is: 876 x 657 pixels
The size of 11110.jpg is: 814 x 610 pixels
The size of 10585.jpg is: 1307 x 980 pixels
The size of 8922.jpg is: 1184 x 888 pixels
The size of 1697.jpg is: 810 x 607 pixels
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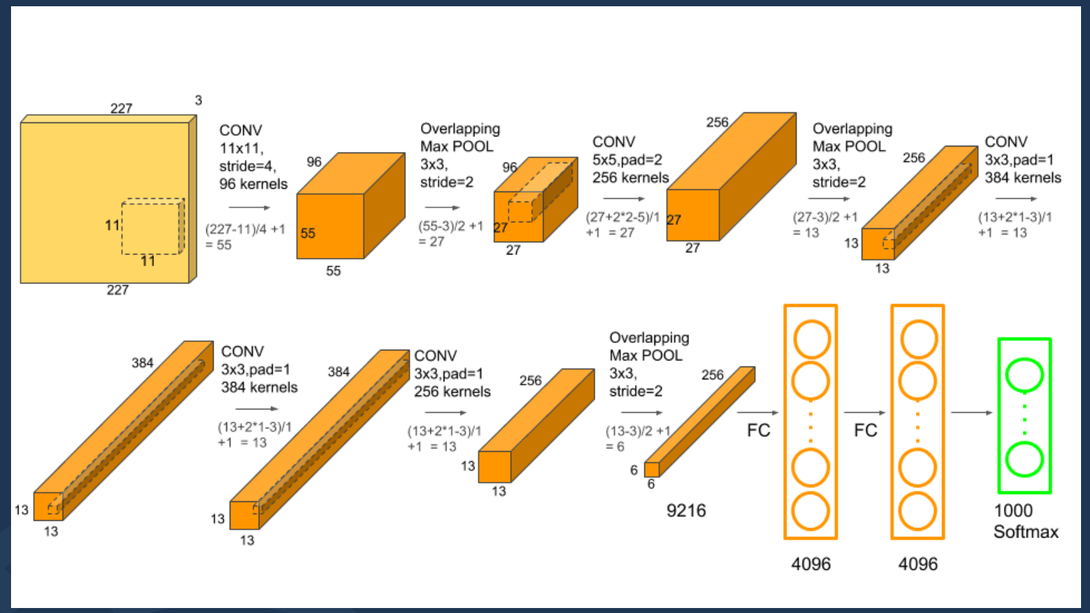
PRE-PROCESSING

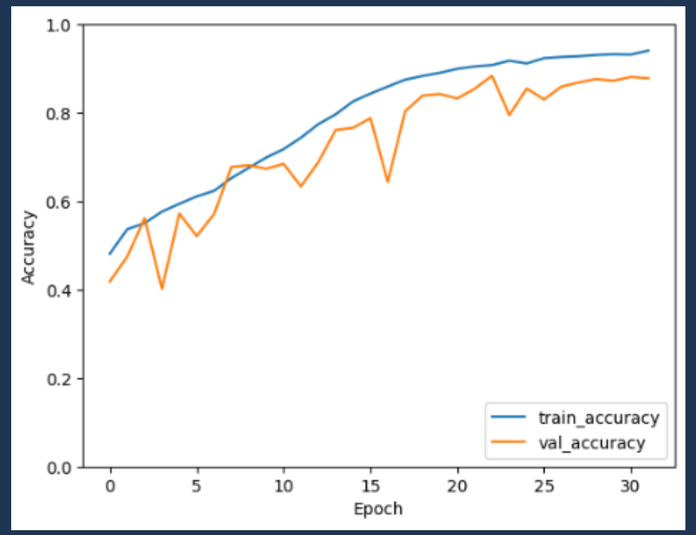
1. Normalisation: Resize Image, LANCZOS
2. Train Test Spilt
3. Data Augmentation
4. Imbalanced Dataset
5. Scalling (/225)

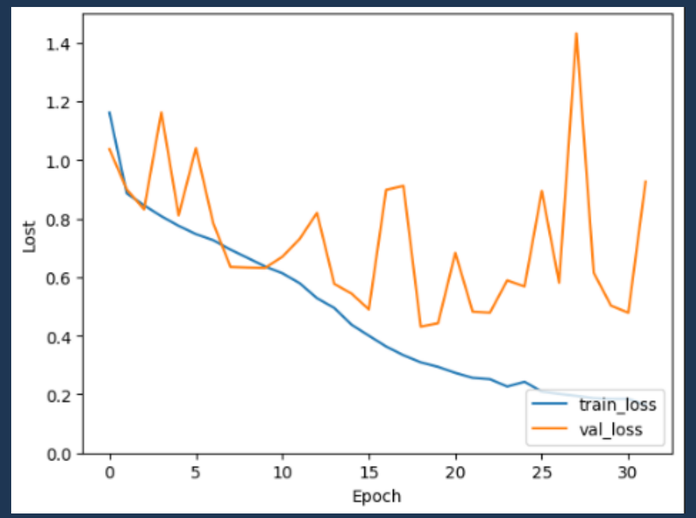




BASELINE MODEL - ALEXNET



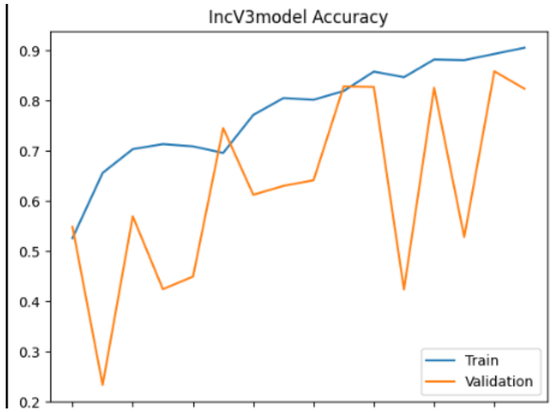


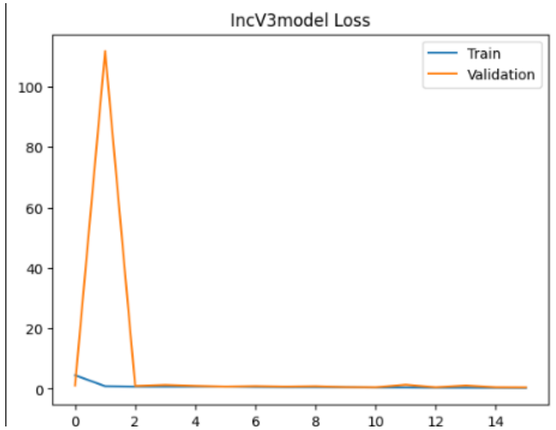


Plans for Improvement:

1. Regularization
2. Early Stopping
3. Transfer Learning
4. Evaluate more metrics

Model Buidling

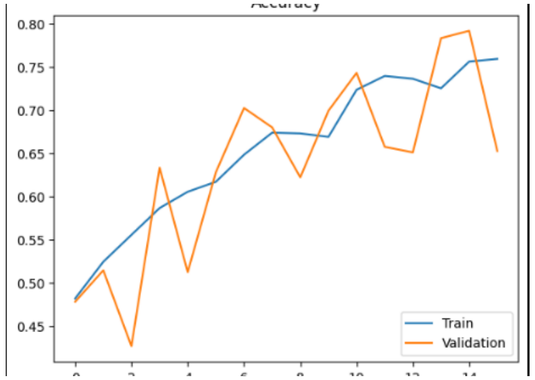


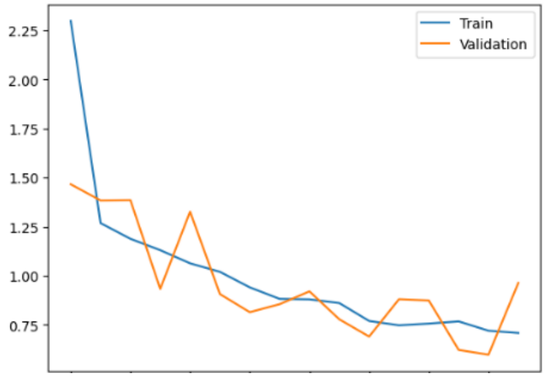


INCEPTION V3

Test Accuracy: 81.90%

Test Loss: 53.15%

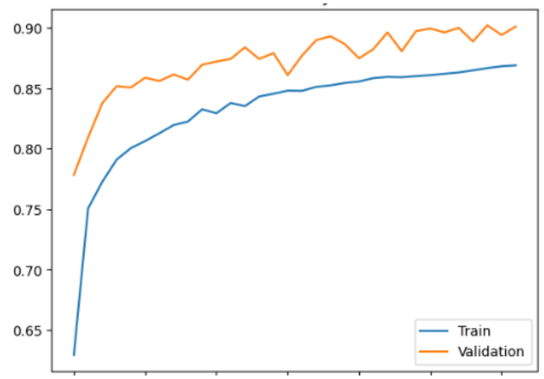


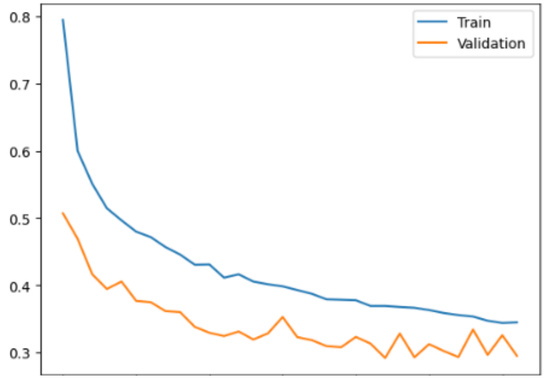


VGGNET 16

Test Accuracy: 64.97%

Test Loss: 95.98%





RESTNET50

Test Accuracy: 90.47%

Test Loss: 31.67%

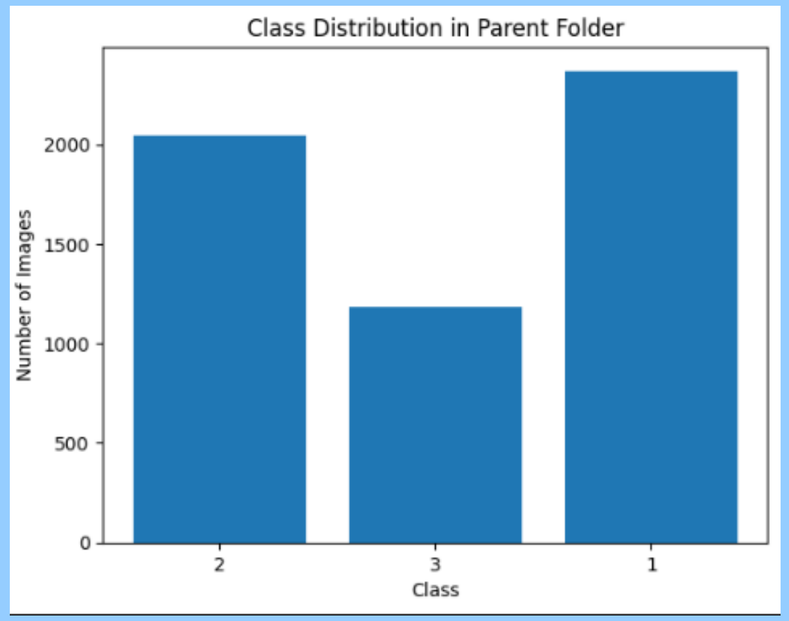


MODEL IMPROVEMENTS 1

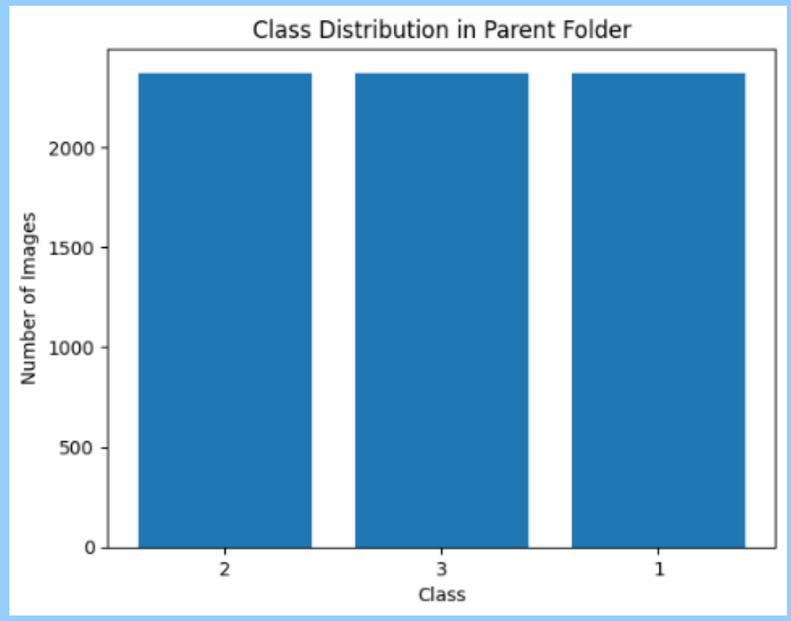
The models show signs of overfitting or underfitting, possibly due to dataset quality issues. Despite no class imbalance, excessive data augmentation might compromise dataset integrity, affecting overall model performance.

Action taken:

1. Remove previously augmented images
2. Downsampling
3. Data Augmentation



BEFORE

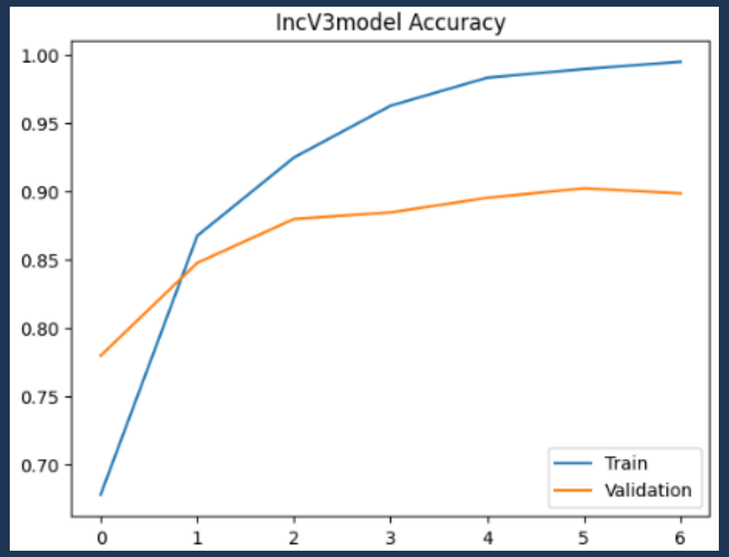


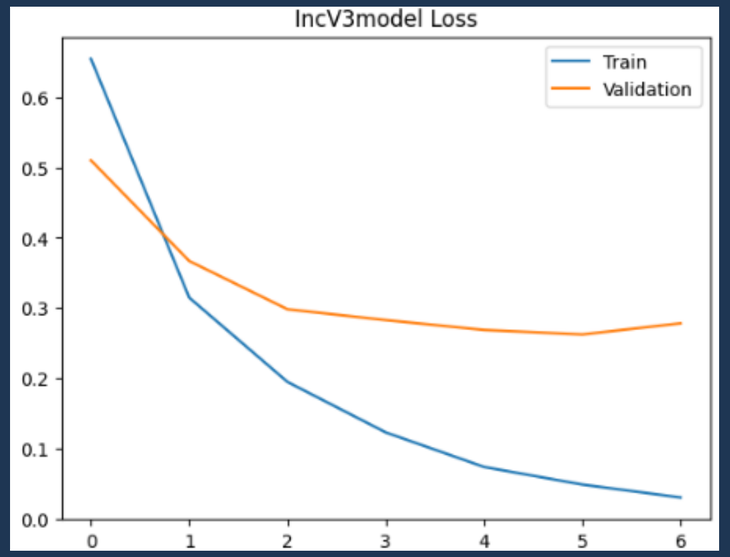
AFTER

MODEL IMPROVEMENTS 2

1. ***Transfer Learning (Pre-Trained Models)***
    - Leads to faster convergence and better generalization, especially when having a limited amount of training data.
2. ***Adjust Learning Rate***
    - Hyperparameter that determines the step size at which the model parameters are updated during training
3. ***Class Weights***
    - To address class imbalance issues by assigning higher weights to underrepresented classes during training
4. ***Early Stopping***
    - A regularization technique that halts the training process when the performance of the model on a validation set starts to degrade or no longer improves

Model Buidling 2

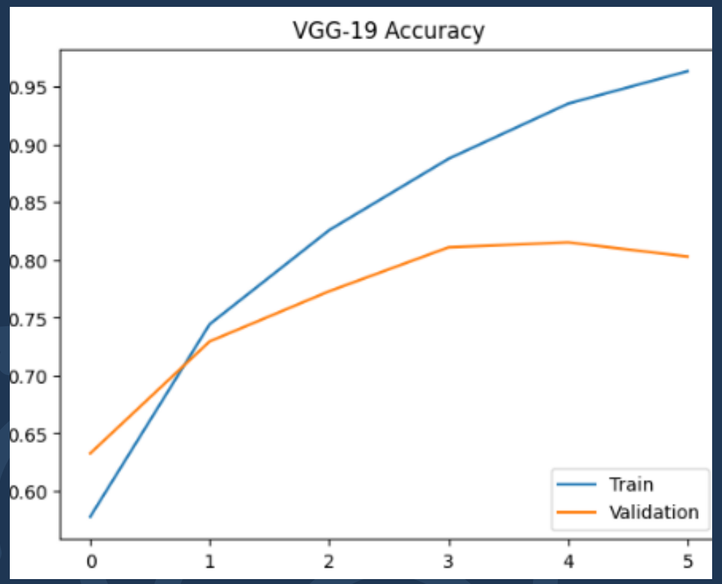


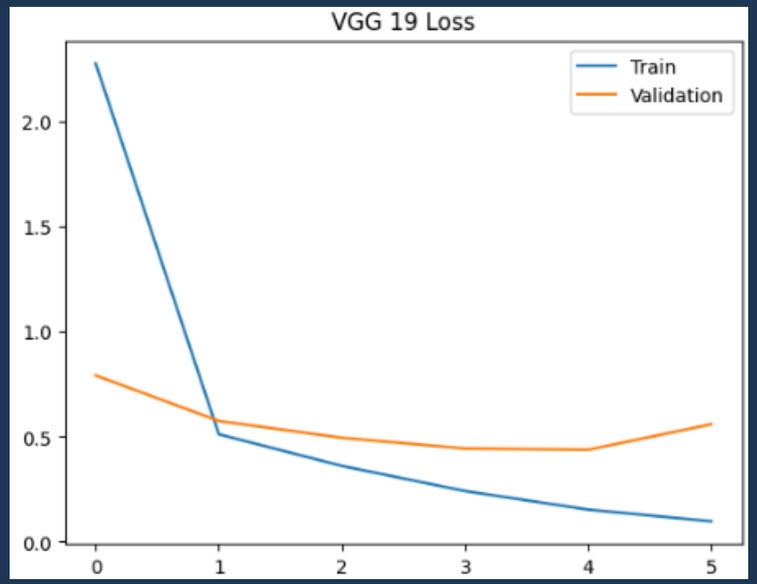


INCEPTION V3

Test Accuracy: 90.68%

Test Loss: 27.37%



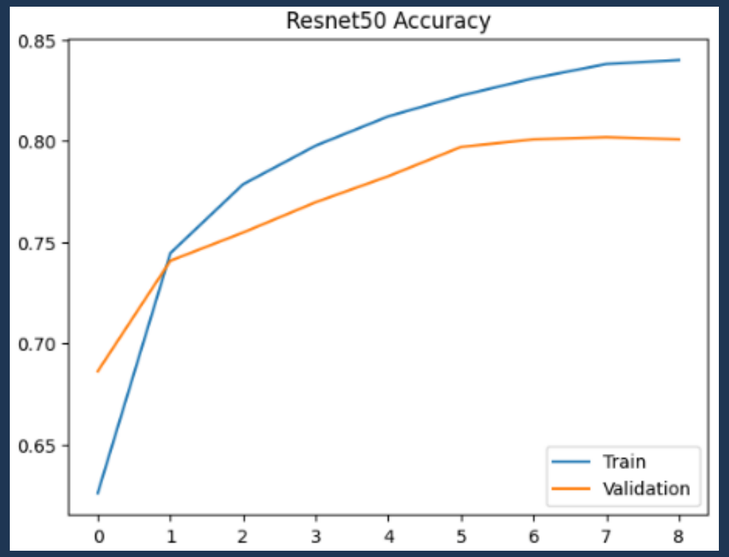


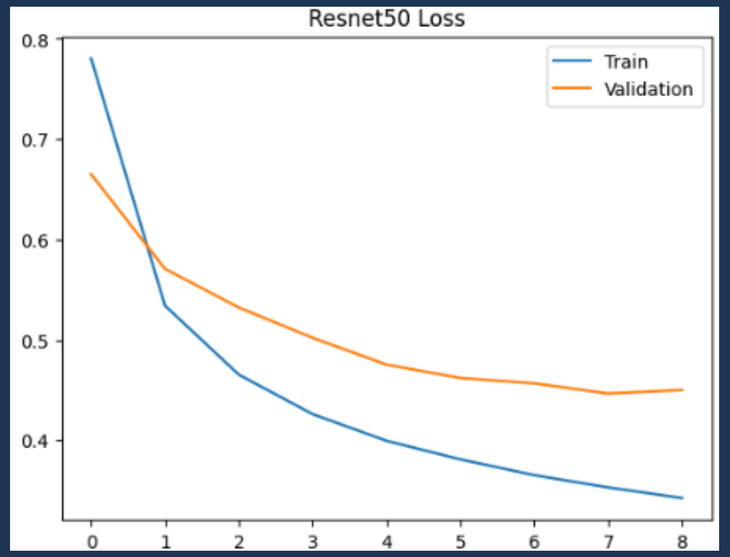
VGGNET 19

Test Accuracy: 81.31%

Test Loss: 45.50%

Model Buidling 2

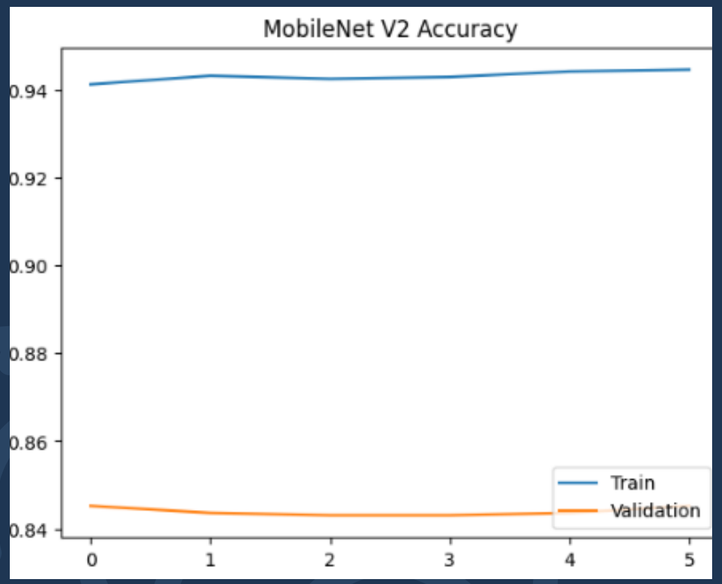


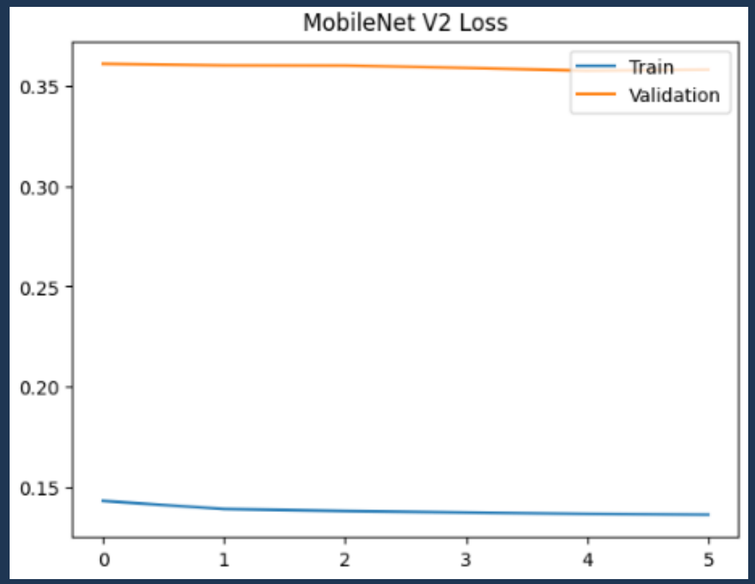


RESNET50

Test Accuracy: 79.70%

Test Loss: 44.72%





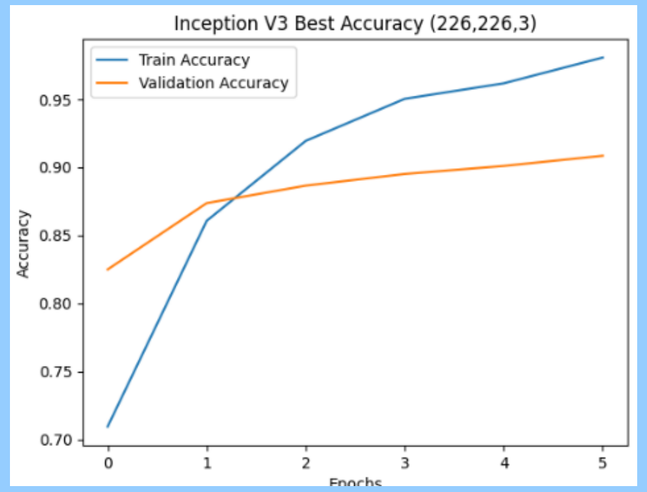
MOBILE NET

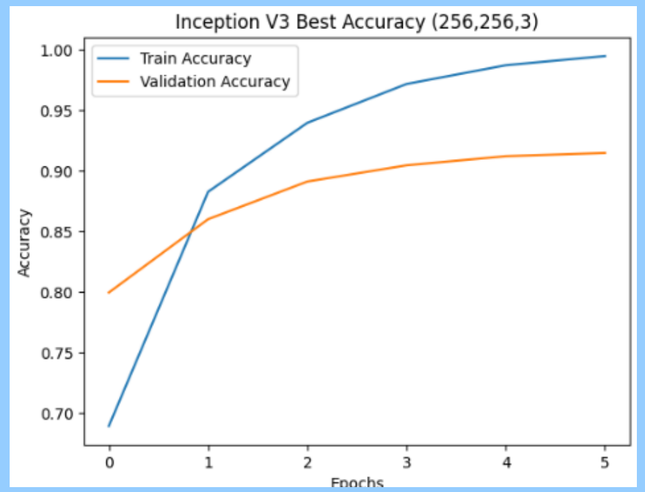
Test Accuracy: 86.23%

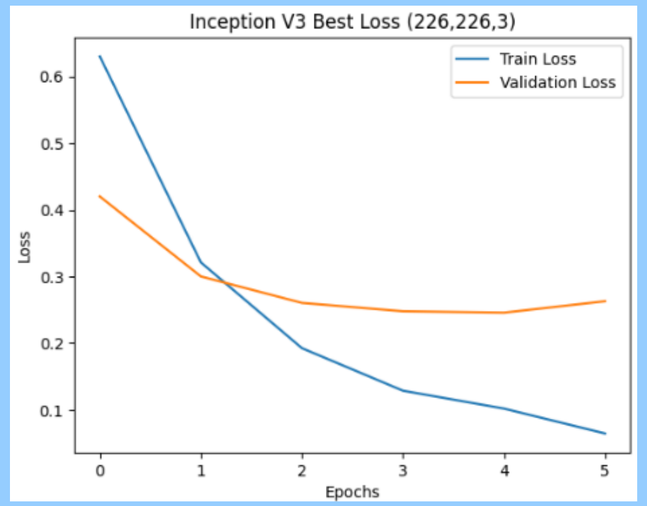
Test Loss: 34.87%

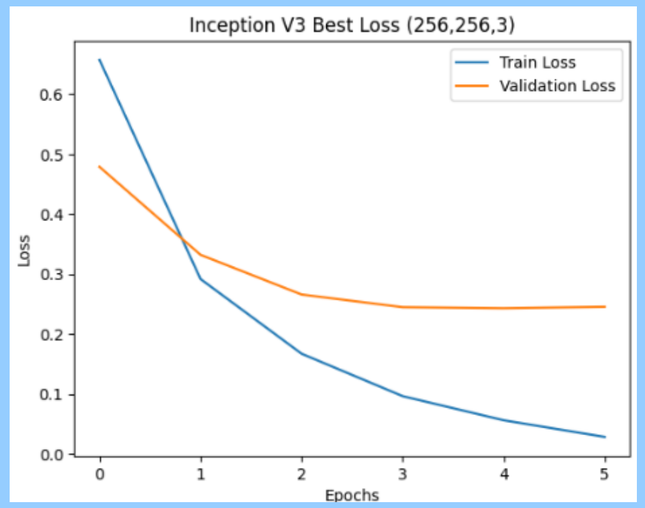
PARAMETERS

Learning rate, Batch size & Input shape









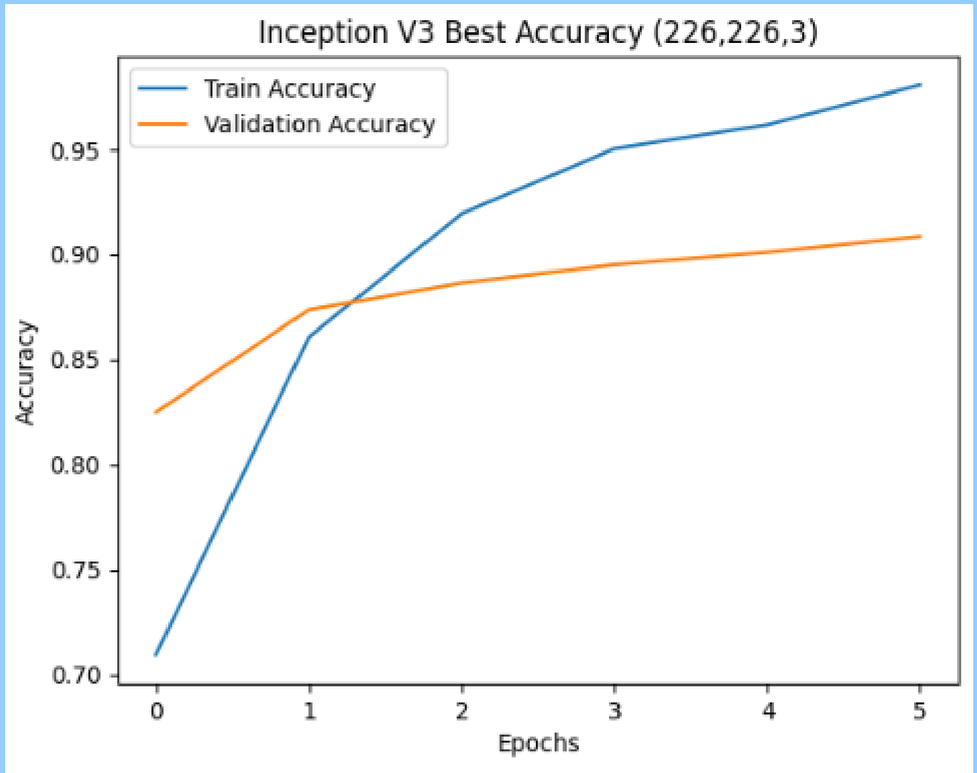
Grid Search

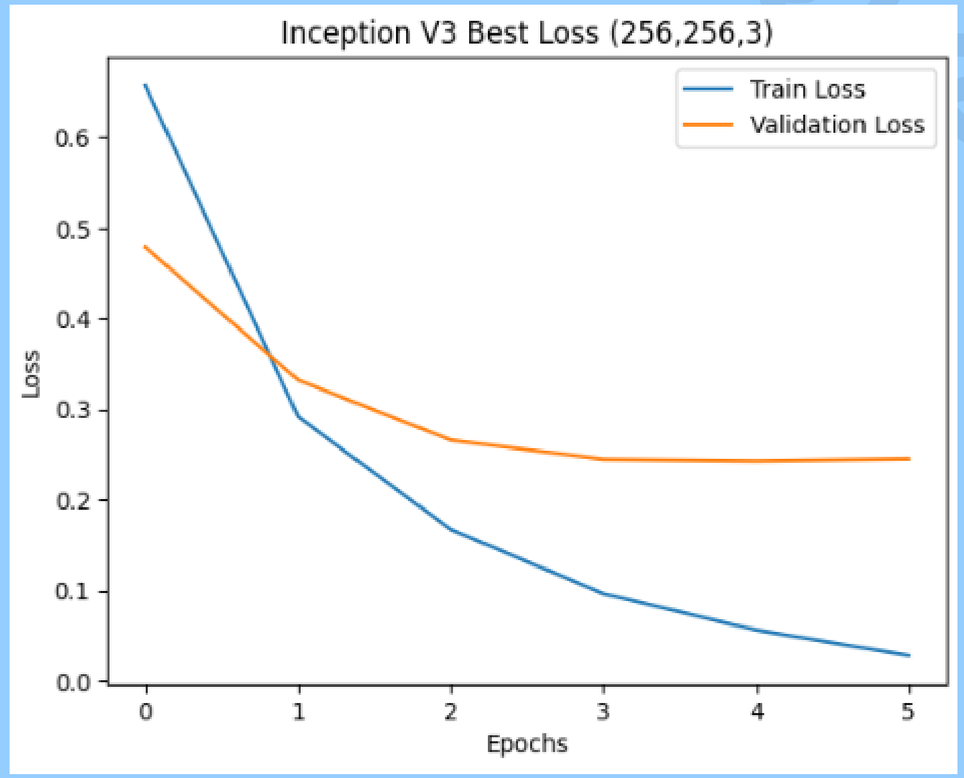
Grid search is a technique used in machine learning to find the optimal hyperparameters for a model.

It works by exhaustively searching through a specified subset of hyperparameter combinations and evaluating each combination using cross-validation.

The goal is to identify the combination of hyperparameters that results in the best performance according to a chosen evaluation metric, such as accuracy or loss.

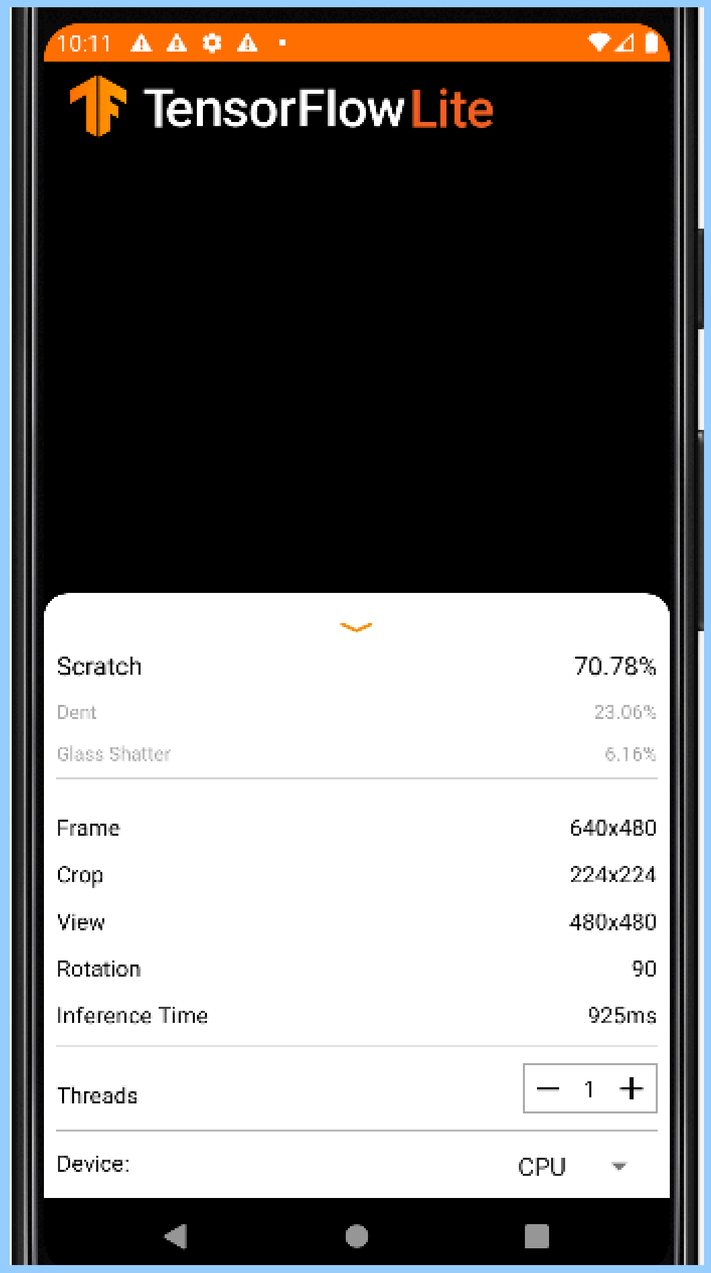
Best Result





By comparison, the best performance is the one with batch size 16, learning rate of 1e-05 and input shape of (226,226,3)

ANDROID DEPLOYMENT



CONCLUSION

Overall, I think this project has helped me to understand and apply the knowledge of deep learning techniques to solve real-world problems.

During the process, I was struggling to get a high accuracy but after much research, I figured that it is due to my over-augmented images.

If given more time, I would have make more changes to my android app to make it more appealing and add one more function to allow user to upload image instead of scanning.